Kathleen Kaye S. Medriano

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SUMMARY

- A 4th year PhD candidate with advanced training in statistics, Bayesian methods and machine learning. Research focuses on developing Bayesian methods for time series data, particularly multi-attribute self-reported mood levels collected through experience sampling methodology (ESM) – received best conference poster award for two years in a row with this research
- Proficient in applying diffusion models to time series data, (e.g. Ornstein-Uhlenbeck); Well-versed in designing amortized inference procedures leveraging neural networks (e.g. time series transformer + conditional invertible neural networks) to enable real-time Bayesian inference; Knowledgeable about causal inference (e.g. causal effect estimation)
- Adept at machine learning methods, including building classifiers, regression models, clustering algorithms, feature engineering, model evaluation, and hyperparameter tuning to enable improved decision-making and drive data-driven strategy
- Internship goals: Contribute to interdisciplinary teams, apply my expertise, and gain exposure to diverse data types to expand my modeling skills and gain insights that will benefit my PhD research while expanding professional networks

EDUCATION

University of California Irvine, Irvine CA, USA (uci.edu) Sep 2022 – Present
Ph.D. in Cognitive Science
M.S. Statistics (acquired Jan 2025), GPA: 3.98/4.00
Relevant courses: Probability Theory and Inference | Statistical Methods (Linear Models, GLMs, Longitudinal data) | Stochastic Processes | Causal Inference | Bayesian Data Analysis | Bayesian Cognitive Modeling | Computational Neuroscience | Introduction to Statistical Learning | Causal Machine Learning | Deep Generative Models

University of the Philippines, Diliman, Philippines (upd.edu.ph) Jan 2017 – Dec 2018
Diploma in Mathematics (with Distinction), GWA: 1.214 (1.0 as highest, 5.0 as lowest)
Honors**:** Top 5% of program graduates, Consistent Dean's Lister (all semesters)

Relevant courses: Mathematical Statistics | Real Analysis | Complex Analysis | Linear Algebra | Abstract Algebra (Groups, Rings, Fields, Graphs) | Numerical Analysis | Operations Research | Optimization and Approximation | Differential Equations (ODEs, intro to PDEs)

University of Santo Tomas, Philippines (ust.edu.ph) Jun 2012 – Jun 2016
B.S. Psychology (Magna Cum Laude), GWA: 1.271 (1.0 as highest, 5.0 as lowest)
Thesis: "'Bahala Na' Attitude (Fatalism) and Psychological Preparedness of Residents living near West Valley Fault"

Honors and scholarship: Magna Cum Laude, Santo Tomas Academic Award Scholarship (Scholarship award for academic excellence; 100% tuition waived throughout whole undergraduate study)**,** Academic Excellence Award in Psychology (valedictorian among psychology graduates), Consistent Dean's Lister (all semesters)

Relevant courses: Test Construction and Measurement | Research Methods | Behavioral Measurement and Analysis

EXPERIENCE

Teaching Assistant Jan 2023 – Present
Department of Cognitive Sciences, University of California, Irvine
- **Lead weekly discussion and lab sessions** for undergraduate courses in cognitive and psychological sciences
- Provided academic support to students through office hours, assignment feedback, and review sessions
- **Courses: *Statistical Inference*, *Exploratory Data Analysis*, *Python Programming*, *Research Methods***

Research Intern Jul 2022 – Sep 2022
Max Planck Institute for Biological Cybernetics (Tübingen, Germany)
- Selected for the **highly competitive CacTüS Internship program** (acceptance rate: < 2%) (cactus)
- Conducted research on **generalization of learned structural knowledge across cognitive tasks using graph similarity** in Dr. Peter Dayan's lab

Independent Researcher Apr 2021 – Feb 2022
Research Collaborations (Manila, Philippines)
- **Conducted statistical analyses (ANOVA, t-tests, etc.) using R** for a study on Filipino attitudes towards LGBTQ+ individuals, focusing on homonegativity and sexual prejudice
- **Collaborated with researchers from The Medical City** (a tertiary hospital) to investigate the **experiences of COVID-19-recovered healthcare workers**, conducting statistical analyses to assess the impact on their well-being and work performance

Benchmark and Portfolio Database Associate Jan 2021 – Apr 2021
Benchmark and Portfolio Database Analyst Apr 2019 – Jan 2021
MSCI (Manila, Philippines)

Worked in a multinational team which is responsible for the processing of index and fund data used by MSCI financial analytics applications.

- **Database Management and Data Quality Assurance**
  - Investigated incidents wherein data quality showed to be subpar; involves performance of calculations and determination of underlying causes (such as incorrect inputs, incorrect financial index methodologies, bug in the code used for processing the data etc.); Data Manipulation Language (DML) and Transaction Control Language (TCL)

- **Code debugging and refactoring, process improvement**
  - Optimized financial data processing codes written in Perl and Python for faster execution and intuitive code reading
  - Wrote a Python script for automatic data processing and issue ticket creation which integrated the Jira REST API

- **Client interfacing**
  - Provided solutions to client inquiries regarding index and fund data; Inquiries range from data quality and accuracy to computational rationale to equity/fixed income index methodology

- **Hiring committee involvement, Training/Mentoring experience, Team Lead Role**
  - After only 4 months into the work, I was already given the authority to interview applicants and provide training to new team members
  - Led the Philippine unit of our multinational financial data services team
  - Responsibilities include facilitating team meetings and overseeing projects of team members

- **Project Management**
  - Spearheaded a major project which involved accommodating a mandatory change of financial industry standard (industry-classification-benchmark-icb)
  - Project centered on optimizing the data processing pipeline of more than 3000 equity indexes to ensure that pipelines remain robust and unaffected by the mandatory change. Tasks involved:
    - o Project planning
    - o Coordinating with different code developers
    - o Project testing and overseeing code deployment to production
    - o Project impact analysis and communication of analysis to clients

Licensed Psychometrician **(Philippines)** Aug 2016
Professional Regulation Commission (PRC), Republic of the Philippines

- Ranked 4th (out of 7,312 examinees) in the 2016 Philippine Board Licensure Exam for Psychometricians (announcement: (prc.gov.ph))

RELEVANT PROJECTS

Reconstruct-then-Bootstrap test for the sufficiency of diffusion process

- **Analyzed the adequacy of diffusion models** for time series satisfying stationarity and Markov properties
- **Utilized theoretical tools** (Kramers-Moyal expansion and Pawula theorem) to assess whether higher-order terms indicate deviations from pure diffusion
- **Developed a bootstrap-based statistical test** to determine when a diffusion model adequately captures observed dynamics of a univariate time series

Bayesian hierarchical Ornstein-Uhlenbeck (BHOU) model for assessment of mobile health (mHealth) psychological intervention

- **Implemented a Bayesian Hierarchical Ornstein-Uhlenbeck (BHOU) model** to analyze time series mood data, assessing the effectiveness of mobile health (mHealth) psychological interventions
- **Computed key evaluation statistics** such as **linear contrasts and Bayes factors** to ensure robust statistical insights
- **Executed the model using JAGS**, leveraging probabilistic programming
- **Analyzed interrupted time series data** collected via randomized controlled trials incorporating Experience Sampling Method (ESM)

Amortized inference of Bayesian Hierarchical Ornstein-Uhlenbeck (BHOU) model

- **Employed amortized inference techniques** for the Bayesian Hierarchical Ornstein-Uhlenbeck (BHOU) model, using simulation-based methods to map data space to model parameter space, enabling rapid and efficient Bayesian inference
- **Designed and experimented with neural network architectures**, including time series transformers, sequence networks, and recurrent neural networks (RNNs), to optimize performance for amortized inference tasks

Bayesian Models for Potential Outcomes Framework

- **Implemented Bayesian models to estimate the causal effect of interventions** within the Potential Outcomes Framework, focusing on Conditional Average Treatment Effect (CATE) estimation
- **Applied the model to real-world data**, analyzing the impact of having additional children on mothers' annual income

Using graph distances to compare Traveling Salesman Problem (TSP) solutions

- **Explored graph-based similarity metrics** to compare Traveling Salesman Problem (TSP) solutions, analyzing features to assess proximity to optimality
- **Applied Multidimensional Scaling (MDS)** to project TSP solutions into a low-dimensional space, uncovering interpretable dimensions that characterize solution properties

Languages: Python, R, MATLAB, Perl, SQL; **Libraries used for projects (for Python):** PyJAGS, PyMC, Arviz, BayesFlow, TensorFlow, PyTorch, NetworkX, nerd, standard data science libraries - Numpy, Pandas, Scikit-learn, Matplotlib, Seaborn etc.,

CONFERENCE PRESENTATIONS and PUBLICATIONS

Medriano, K., Oravecz, Z., & Vandekerckhove, J. (2025, July). Data-driven model discovery for coupled time series. Abstract published at MathPsych / ICCM 2025. Via mathpsych.org/presentation/1924. **(TALK)**

Medriano, K., Oravecz, Z., & Vandekerckhove, J. (2024, July). Amortized stochastic time series models for intervention designs. Abstract published at MathPsych / ICCM 2024. Via mathpsych.org/presentation/1633. **(BEST POSTER)**

Medriano, K., Oravecz, Z., & Vandekerckhove, J. (2023, July). Bayesian hierarchical Ornstein-Uhlenbeck models for intervention designs. Abstract published at MathPsych/ICCM/EMPG 2023. Via mathpsych.org/presentation/1235. **(BEST POSTER)**

Etz, A., Chávez de la Peña, A. F., Baroja, L., **Medriano, K.**, & Vandekerckhove, J. (2024). The HDI+ROPE decision rule is logically incoherent but we can fix it. Psychological Methods.

Carascal, M. B., Capistrano, P. E., Figueras, M. D. L., Cataylo, O. L. A. C., Zuñiga, S. M. S., Reyes, M. E. S., **Medriano, K. K. S.,** Gamo, A. T., Mendoza, P. D., & Macalipay, S. L. B. (2022). Experiences of COVID-19-Recovered Healthcare Workers in a Tertiary Hospital in the Philippines: A Mixed-Method Inquiry. Inquiry : a journal of medical care organization, provision and financing, 59, 469580221107051. https://doi.org/10.1177/00469580221107051

Reyes, M.E.S., Davis, R.D., Salonga, B.M., Cheng, H.A.T., Vasquez, A.N.V., Cruz, G.J.S., Muslim, N.M., **Medriano, K.K.S.**, & McCutcheon, L.E., (2021). Sexual prejudice: Filipinos' attitudes toward lesbians and gay men. North American Journal of Psychology, 23(4), 617-638

HONORS, CERTIFICATIONS AND SCHOLARSHIPS

Tarow and Minako Indow Fellowship (2025) Oct 2025
Department of Cognitive Sciences, University of California, Irvine

- Fellowship awarded annually to two graduate students in recognition of exceptional research skills and productivity, as demonstrated during advancement to PhD candidacy
- Awarded for research on 'Methods for Time Series Data in Cognitive and Psychological Sciences' under advisor Joachim Vandekerckhove.

Magna Cum Laude Jun 2016
College of Science, University of Santo Tomas, Philippines

- Academic distinction for graduating with honor

Academic Excellence Award in Psychology Jun 2016
College of Science, University of Santo Tomas, Philippines

- Awarded for graduating as the top student in my academic program

Academic Program Award Jun 2016
UST Psychological Alumni Association, Philippines

- Awarded for exhibiting exemplary academic performance

Santo Tomas Academic Award Scholarship Jun 2012 – Jun 2016
University of Santo Tomas, Philippines

- Scholarship award for academic excellence; 100% tuition waived throughout whole undergraduate study